October 2026

Dear Editor-in-Chief and Editors,

Please find enclosed the manuscript **"Streaming LLM Weights from CXL Memory: A Simulation Study of Regimes, Placement and Pipeline-Aware Pinning"** for consideration in the *Microprocessors and Microsystems: Embedded Hardware Design*. The study has been rebuilt from an earlier version with a new simulator, stronger baselines and new experiments, and is presented explicitly as a simulation and analysis study.

Problem. Models such as Llama-3-70B (141 GB in FP16) exceed GPU memory. CXL Type-3 memory expanders offer cheap host-attached capacity, but a GPU cannot address them: weights reach HBM through host-initiated DMA at roughly 10–20 GB/s on published prototypes. We ask when streaming weights over this path is viable and what a runtime policy can still gain.

Contributions. (1) A host-mediated transfer model, a closed-form arithmetic-intensity threshold, an implementation-independent lower bound and a regime map: streaming is nearly free only above about 16k tokens per iteration and tens of times slower than resident weights for small-batch decode. (2) An analysis of where policy gains come from: placement of the resident layers matters when compute and transfer are comparable (contiguous sets idle the link), and a tuned FlexGen-style per-layer split is already within a few per cent of our policy on dense uniform models. (3) CAMP, a runtime policy (work-conserving prefetcher with next-use admission and stall-cost-aware pinning) that stays within 1.07 of the lower bound in simulation, is up to 2.8× faster than demand paging, ties the best baseline on dense models and is ahead where unit sizes or reuse are non-uniform (up to 1.97× on encoder–decoder inference). (4) A pool-sizing rule: when G GPUs share one expander the threshold rises G-fold, so streaming stays nearly free while $G \leq T/T^{\star}$ (confirmed in simulation up to 16 GPUs). (5) Negative results: the compute-time estimate changes latency by under 1% once calibrated, and LFU is indistinguishable from LRU on cyclic scans. (6) A simulator, cross-checked against a published CXL weight-streaming result, whose link and overlap logic is checked on a real GPU, with the accuracy of that check reported as it is (mixed: 0.6% in link-bound configurations, up to 25% elsewhere).

Fit with the journal. The journal covers the memory and communication structures of complex processor platforms and the interplay between hardware and software with emphasis on hardware. The manuscript studies one such structure: a host-mediated path through which an accelerator obtains data from a CXL-attached memory tier. It models the path's bandwidth, per-transfer costs and overlap with computation, derives when the path can be hidden behind computation, and evaluates a data-movement scheduler for it. We are aware that the journal's focus is embedded hardware and that this is a platform-level study of an accelerator memory hierarchy rather than a circuit or SoC design; we believe the memory-system and co-design questions it addresses are of interest to the readership.

Scope of the evidence. This is a simulation study and the manuscript says so in its title, abstract and introduction. No CXL hardware was measured; the CXL link is simulated with parameters anchored in published measurements and the pipeline logic is validated on a PCIe-attached GPU with pinned host memory standing in for the pool. Mixture-of-Experts models, multi-GPU or multi-tenant pools and CXL switches are not evaluated, and the T4 check leaves part of its own error unexplained. Code, raw results and the hardware micro-benchmark are publicly available (see the Data availability statement).

The manuscript is original, is not under consideration elsewhere, and the authors declare no compet-

ing interests. The generative-AI declaration required by the journal is included in the manuscript.

Sincerely,

Quang-Vinh Dang (corresponding author), Dat Le, Minh Ngoc Dinh
British University Vietnam, Hung Yen, Vietnam; Ho Chi Minh City University of Economics and Finance; Millennia Education, Ho Chi Minh City, Vietnam
vinh.dq4@buv.edu.vn ORCID 0000-0002-3877-8024